themselves only beyond the range of our current observational capacity, a circular shape is consistent with the universe we observe. But for dimensions that stay small, a more realistic scenario is one in which they are curled up into a more intricate Calabi-Yau space. Of course, the key question is, Which Calabi-Yau space? How is this particular space determined? No one has been able to answer this question. But by combining the drastic topology-changing results described in the preceding chapter with these cosmological insights, we can suggest a framework for doing so. Through the space-tearing conifold transitions, we now know that any Calabi-Yau shape can evolve into any other. So, we can imagine that in the tumultuous, hot moments after the bang, the curled-up Calabi-Yau component of space stays small, but goes through a frenetic dance in which its fabric rips apart and reconnects over and over again, rapidly taking us through a long sequence of different Calabi-Yau shapes. As the universe cools and three of the spatial dimensions get large, the transitions from one Calabi-Yau to another slow down, with the extra dimensions ultimately settling into a Calabi-Yau shape that, optimistically, gives rise to the physical features we observe in the world around us. The challenge facing physicists is to understand, in detail, the evolution of the Calabi-Yau component of space so that its present form can be predicted from theoretical principles. With the newfound ability of one Calabi-Yau to change smoothly into another, we see that the issue of selecting one, Calabi-Yau shape from the many may in fact be reduced to a problem of cosmology.5

Before the Beginning?

Lacking the exact equations of string theory, Brandenberger and Vafa were forced to make numerous approximations and assumptions in their cosmological studies. As Vafa recently said,

Our work highlights the new way in which string theory allows us to start addressing persistent problems in the standard approach to cosmology. We see, for example, that the whole notion of an initial singularity may be completely avoided by string theory. But, because of difficulties in performing fully trustworthy calculations in such extreme situations with our present understanding of string theory, our work only provides a first look into string cosmology, and is very far from the final word.6